

Local Binary Pattern

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Defaults

```
clc;  
clear;  
close all;
```

Inputs, Variables, Constants

```
img = imread('Javanese-Cat.jpeg');  
img = rgb2gray(img);  
img = double(img);  
  
[rows, cols] = size(img);
```

LBP

```
LBP = zeros(rows-2, cols-2);  
  
% LBP calculation  
for i = 2:rows-1  
    for j = 2:cols-1  
  
        center = img(i,j);  
  
        %neighbours comparison  
        code = [ img(i-1,j-1) >= center,...  
                 img(i-1,j)   >= center,...  
                 img(i-1,j+1) >= center,...  
                 img(i,j+1)   >= center,...  
                 img(i+1,j+1) >= center,...  
                 img(i+1,j)   >= center,...  
                 img(i+1,j-1) >= center,...  
                 img(i,j-1)   >= center ];  
  
        % converting pattern into decimal  
        weights = [1 2 4 8 16 32 64 128];  
        LBP(i-1,j-1) = sum(code .* weights);  
  
    end  
end
```

```
LBP = uint8(LBP);
```

Plots

```
figure
subplot(1,2,1)
imshow(uint8(img))
title('Original Image')

subplot(1,2,2)
imshow(LBP)
title('LBP Image')
```

